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ORIGINAL ARTICLE

Machine learning helps predict long-term mortality and graft failure in patients undergoing heart transplant

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Abstract

Objective We aimed to develop a risk prediction model using a machine learning to predict survival and graft failure (GF) 5 years after orthotopic heart transplant (OHT).

Methods Using the International Society of Heart and Lung Transplant (ISHLT) registry data, we analyzed 15,236 patients who underwent OHT from January 2005 to December 2009. 342 variables were extracted and used to develop a risk prediction model utilizing a gradient-boosted machine (GBM) model to predict the risk of GF and mortality 5 years after hospital discharge. After excluding variables missing at least 50% of the observations and variables with near zero variance, 87 variables were included in the GBM model. Ten fold cross-validation repeated 5 times was used to estimate the model's external performance and optimize the hyperparameters simultaneously. Area under the receiver operator characteristic curve (AUC) for the GBM model was calculated for survival and GF 5 years post-OHT.

Results The median duration of follow-up was 5 years. The mortality and GF 5 years post-OHT were 27.3% ($n=4161$) and 28.1% ($n=4276$), respectively. The AUC to predict 5-year mortality and GF is 0.717 (95% CI 0.696–0.737) and 0.716 (95% CI 0.696–0.736), respectively. Length of stay, recipient and donor age, recipient and donor body mass index, and ischemic time had the highest relative influence in predicting 5-year mortality and graft failure.

Conclusion The GBM model has a good accuracy to predict 5-year mortality and graft failure post-OHT.

Keywords Heart transplant · Machine learning · Mortality · Graft failure

Introduction

Orthotopic heart transplantation (OHT) is currently the definitive treatment of severely advanced heart failure not responsive to the appropriate medical therapy, resynchronization therapy, and/or valve repair. Mechanical circulatory support devices have shown promise in select patients but, unlike OHT, their associated survival and complication rates have not been studied long term [1]. Meanwhile, the number of OHTs has been increasing across the globe, reaching 4388 adult OHTs at 285 centers in 2015. Most significant increases have been found in North America [2]. In 2015, the United States totaled 2804 OHTs and then climbed further to about 3200 OHTs in 2016, though there are over 4000 patients still on the waitlist as of December 2017 [3, 4]. The supply of heart donors has not risen to match demand, placing greater attention towards appropriate organ allocation in OHT. A standardized, highly accurate model needs to be established to help guide clinicians with estimated

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short- and long-term mortality risk based on a composite of many donor and recipient pre-, peri-, and post-operative characteristics.

The recent 2014 report by the International Society of Heart and Lung Transplantation (ISHLT) studied almost 4000 adults at over 400 transplant centers, showing a survival rate post-transplantation of 84.5% at 1-year and 72.5% at 5-years [1]. The most frequent causes of death in the first year include acute graft failure (within the first 30 days), infection, and multiple organ failure. Complications such as renal failure, cardiac allogeneic vasculopathy, and malignancy become significant at 5 and 10 years after OHT [5]. While improvements in immunosuppression therapy have reduced graft rejection, management choices are complicated by the increased risk of renal failure with the commonly used immunosuppressants such as tacrolimus, a calcineurin inhibitor [1]. Postprocedural management choices can alter the morbidity and mortality risk and can be included in the model along with other patient characteristics.

Previous studies by ISHLT using multivariable analysis have shown mortality and complication rates vary based on specific patient and donor pre-procedural factors. Some main contributory factors associated with unfavorable mortality hazard ratios include older age of donor, older age of recipient, extremely high or low weight of recipient, elevated recipient creatinine, re-transplant recipient, recipient history of temporary circulatory support, dialysis, and ventilator support immediately prior to transplant surgery, among multiple others. Favorable hazard ratios for recipient survival were seen with recipient being an insulin-dependent diabetic and recipient not having been hospitalized just prior to transplant surgery, this was reported in the 30-s official adult heart transplantation report in 2015 [5]. Furthermore, different recipient diseases carry different risk profiles, with best outcomes at 1 year seen in cardiomyopathy recipients and best outcomes at 5 years seen in congenital heart disease recipients [5]. Since short- and long-term survival and complication rates have been shown in international studies to vary based on different baseline characteristics of both recipients and donors, a standardized mortality risk algorithm addressing multiple risk factors together can help estimate composite mortality risk.

OHT models have already been constructed; however, they are either limited in duration of prediction, based on very few factors, or not strong enough to be clinically useful. The 2011 Recipient Risk Index for Mortality after Cardiac Transplantation (IMPACT) study created a 50-point scoring system based on only 12 specific recipient variables which showed accurately predicted survival rates in the validation cohort at 1-year. Survival rates varied based on scores, with lower scores reflecting lower risk: 0–2 (92.5%); 3–5 (89.9%); 7–9 (86.3%); and 10 or greater (74.9%); $p < 0.001$ [6]. The 2015 International Heart Transplant Survival Algorithm (IHTSA) has been shown to be highly accurate with predicted versus actual 1-year, 5-year, and 10-year survival rates of 84.5% versus 84.4%, 72.9% versus 75.6%, and 57.5% versus 57.5% in the external validation cohort; however, this study's c-index of 0.60 reflects poor predictability [7]. Predictability may be improved by including more extensive, updated recipient and donor information and by employing a more powerful analysis tool, machine learning.

Machine learning has improved the accuracy of predicting cancer susceptibility, recurrence, and mortality by 15–25% in high-quality studies [8]. Although highly predictive, the machine learning method requires large, robust data sets [8]. The International Society of Heart and Lung Transplant (ISHLT) Registry was selected as a data source because it is the world's current largest registry of OHTs, including data from 223 centers in 18 countries since 1983 (https://www.ishlt.org/registries). Furthermore, it collects data on almost 400 variables before, during, and after transplantation until death, including morbidity and mortality, with established quality control measures to minimize inclusion of poor data [7].

Such a model for OHT predicting future outcomes based on multiple individual donor and recipient variables can help guide clinician recommendations and patient decisions regarding OHT, as well as the post-transplant management. The objective of this study is to apply machine learning algorithms to predict all-cause mortality and graft failure at 5 years after OHT.

Methods

Data source

We obtained data from the ISHLT Registry, the world's current largest registry of OHTs, including data from 223 centers in 18 countries since 1983. Furthermore, it collects data on almost 400 variables before, during, and after transplantation until death, including morbidity and mortality, with established quality control measures to minimize inclusion of poor data. The ISHLT data are collected via manual entry, electronic download from individual OHT centers, or via data sharing with national/regional organ procurements organization or organ exchange organizations. Post-transplant data are reported annually until death, including date of death, for each OHT. A complete list of all OHT centers participating in the ISHLT registry can be found at the ISHLT website (https://www.ishlt.org/registries).

Study population

Data on OHT donors and corresponding recipients were collected from January 2000 to June 2017 (Supplementary data) in the ISHLT registry, yielding 78,852 patients. When filtering for age > / = 18, 64,442 remain and when excluding patients with multiple organ or previous transplants ($n = 3181$), 61,261 remain; with BMI < 15 ($n = 168$), 61,093 remain; with missing status ($n = 33$), 61,060 remain; with OHT outside the window 2005–2009 ($n = 44{,}367$), 16,693 remain; with sudden death ($n = 1451$), 15,242 remain; with donor gender missing ($n = 2$), 15,240 remain; and finally with time of graft failure/death missing ($n = 4$), 15,236 remain, the final study population included 15,236 OHT patients. A diagram is provided in supplementary data showing the above cohort derivation. The last annual follow-up data were collected on December 2009.

In addition to the filtering above, we filtered out patients with a missing value for time (i.e. survival time, graft survival time) prior to modeling each outcome and, therefore, the number of observations varies by outcome.

Inclusion and exclusion criteria

Inclusion criteria:

1. All patients > = 18 years, who underwent orthotopic heart transplant with follow-up data available at 5 years.

Exclusion criteria:

1. Patients who underwent heart/lung transplant.
2. Patients who underwent heart/kidney transplant.
3. Patients with OHT prior to 2005 and after 2009.
4. Patients with history of prior heart transplant.

Study design

The two primary endpoints are all cause mortality and graft failure at 5 years after OHT. Predictability of endpoints based on both donor and recipient variables were evaluated using the gradient-boosted machine (GBM) algorithm [9, 10].

The study population data set was divided into a 'training cohort' in which the mortality risk algorithms were derived and a 'validation cohort' in which the algorithms were applied and tested. The training cohort was derived from a random sample of 80% of extracted data and the validation cohort comprised the remaining 20%. The training cohort was used to create the predictive models utilizing tenfold cross-validation repeated 5 times to tune the GBM hyper parameters (number of trees, interaction depth, and shrinkage). The minimum number of observations required in each node, which is the point in a tree where a classification or split is made, was fixed at 20. The validation cohort was then used to evaluate the model using AUC curves.

The variables gathered were the demographic and transplant characteristics of the donors and recipients (342 variables) from January 2005 to March 2009. Variables missing more than 50% of observations were then removed. Blank categories for categorical variables were recoded as 'No' or a 'No' equivalent and some variables such as functional status were recoded to have fewer and, as a result, less sparse categories. Lastly, variables with near-zero variance (e.g. variables where a single category comprised 95% or more of the information in a variable) were removed. 87 variables made it to the modeling phase.

Statistical analysis

Each survival outcome was treated as a traditional binary outcome variable and time-to-event was not taken into account. This requires that patients had a patient's procedure occurred between 2005 and 2009 so that all patients had at least 5 years post-OHT. Patients who were lost to follow-up prior to year 5 were treated as if they had not experienced the event of interest (e.g. mortality or graft failure). The *caret* R package [11] was used to fit a gradient-boosted machine (GBM) from the *gbm* R package [12] using tenfold cross-validation repeated 5 times to optimize the model hyperparameters, where hyperparameters are variables that configure the model to run or behave a certain way and need to be defined prior to the model being fitted on the training data. The optimal value for the hyper parameters was defined as the combination of values that resulted in the highest area under the receiver operating characteristic curve (AUC), which is a numeric representation of the model's ability to discriminate between patients with events and those without events at a pre-specified time point.

The efficacy of the predictive model was then determined on the holdout set which comprised of 20% of the 15,236 observations that met our entry criteria. Sensitivity, specificity, positive predictive value, and negative predictive value along with 95% normal approximated confidence intervals are provided in addition to AUC to quantify the final model's predictive ability. The 95% confidence interval for AUC was calculated using the DeLong method [13] ROC curves are produced using the *pROC* R package [14].

Continuous variables were summarized by mean, standard deviation, and range while categorical variables were summarized by count and percent. Continuous variables were compared between the derivation and validation data sets using the equal-variance *t* test, while the Chi-square test was used to compare the distribution of categorical variables. All hypothesis tests were two-sided with $p < 0.05$

considered statistically significant. There was no adjustment for multiple testing. All analysis was performed in R 3.4.2.

Results

Characteristics of the derivation and validation cohorts

The derivation and validation cohorts comprised 15,236 OHT patients with a median duration of follow-up of 5 years (Q1 2.2 years, Q3 5 years). The overall 5-year mortality after OHT was 27.3% ($n = 4161$). The derivation cohort (DC) and validation cohort (VC) were similar with regard to the majority of measured variables (86 independent variables). Variables with significant difference between the two groups included preoperative functional status (DC higher average functional status than VC $p = 0.019$), inotrope use (higher in VC $p = 0.043$), gender (more males in DC $p = 0.021$), treatment with basilixumab for induction therapy (more in VC $p = 0.040$), donor total bilirubin (higher in VC $p = 0.006$). Comparison of the two groups by the variables measured is presented in Supplementary data.

Derivation of the survival model

The final model included 87 variables resulting in an area under the ROC curve (AUC) for prediction of mortality and graft failure at 5 years of 0.717 (95% CI 0.696–0.737) and 0.716 (95% CI 0.696–0.736) respectively (Fig. 1). Of the 15,236 patients in total, 1,524 received a unique survival time prediction with our model. Of the 87 variables included in our model, ten showed significant independent influence on the two outcomes. The variables with highest relative influence (RI) were, from most relative influence to least, in the graft failure group: hospital length of stay (RI 21.1), recipient age (RI 7.5), immunosuppressive medications given for maintenance or anti-rejection at time of discharge (RI 6.3), ischemic time (RI 5.0), donor age (RI 4.9), graft function at discharge (RI 4.4), recipient BMI (RI 3.9), donor BMI (RI 3.5), pre-transplant creatinine (RI 2.9), and pre-transplant bilirubin (RI 2.8). Similarly, for the 5-year mortality group, from most relative influence variable to least: hospital length of stay (RI 21.0), recipient age (RI 7.4), ischemic time (RI 6.5), donor age (RI 6.0), donor BMI (RI 4.9), recipient BMI (RI 4.6), immunosuppressive medications given for maintenance or anti-rejection at time of discharge (RI 4.5), graft function at discharge (RI 3.8), pre-transplant bilirubin (RI 3.0), pre-transplant creatinine (RI 2.5). The effect on graft failure and mortality was different, however, for each variable, though each variable's effect reached significance. This information is displayed in supplementary data.

Sub-group analysis

As illustrated in Fig. 2, a subgroup analysis was performed for the top four variables influencing the model to predict both 5-year graft failure and 5-year mortality. These four variables were recipient age, ischemic time, length of hospital stay, and donor age. Evaluation of these predictors shows that the model was better at predicting graft failure at 5 years if the recipient was > 60 years of age compared to all other age groups ($p < 0.004$). The model was more accurate in predicting 5-year mortality if the ischemic time was > 4 h ($p < 0.007$). No differences existed in predictive power of the model based on subgroups of donor age, or length of stay for either graft failure or mortality.

Discussion

In this paper, we present a powerful analysis tool for predictive evaluation of patients prior to OHT using a complex machine learning algorithm. This analysis included more extensive, updated recipient, and donor information as compared with previous models [6, 7, 15–18]. In addition, this model was used to not only predict long-term mortality after

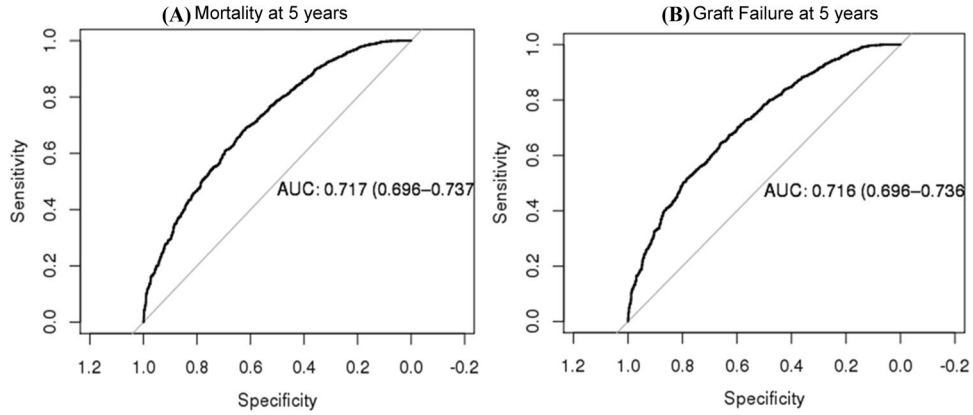


Fig. 1 Receiver operating characteristic and area under the curve for prediction of **a** mortality and **b** graft failure at 5 years

OHT, but also used to predict the likelihood of graft failure after 5 years, another clinically significant outcome in this patient population which has not been as extensively studied in previous models [15]. Definition of graft failure in this study is the same as used for reporting to the ISHLT database; that is: any cause of decompensated failure of the LV or RV or both due to any of the following: primary failure in the immediate postoperative period, hyperacute, acute or chronic rejection, graft infection, recurrent disease, chronic allograft vasculopathy (CAV) or any other cause [5]. Use of this new model can accurately predict graft failure and 5-year mortality prior to transplant based on hundreds of pre- and post-transplant variables. This study further solidifies the notion that non-linear ANN models can be used to predict mortality and outcomes with high accuracy [8].

In terms of predictive strength and clinical importance, the AUC is the best tool for describing a model's performance [7, 19, 20]. Our model, with an AUC of 0.717 and 0.716 for 5-year mortality and graft failure, respectively, predicts outcomes with great accuracy and improves on all previous models of estimating post-transplant success. In addition, this model included more variables and updated recipient and donor information as previously mentioned, giving it greater predictive value overall. Previous models, with the exception of the recent IHTSA model, have attempted to predict short-term outcomes, namely 1-year mortality. Our model has extended on these previous algorithms with high-accuracy outcome prediction for mortality and graft failure 5 years after OHT.

The most important variable for predicting mortality and graft failure after 5 years in our model was hospital length of stay, which differs from previous studies in which donor age was reported as being the most significant factor [6, 7, 21]. Increasing length of hospital stay increased the risk of 5-year mortality and graft failure about three times more than any other variable measured in the study. Donor age, previously reported to be the most important variable in predicting post-transplant mortality, was also shown to be a significant variable in predicting mortality and graft failure in our model; however, this particular variable was found to have less relative influence than several others discussed below namely: recipient age and ischemic time. Due to the fact that hospital length of stay is difficult to modify and cannot be predicted prior to transplant, other variables of importance were more significant for selection criterion to maximize transplant success prior to allograft according to our model. However, any variable known to affect the length of a recipient's stay should be optimized to increase the likelihood of graft and patient survival at 5 years.

Second to length of stay, in terms of predictive effect, was recipient age prior to transplant. This finding is unsurprising given that the older a patient is before transplant the less likely they are to survive after 5 years and the more comorbidities the patient is likely to have. In previous studies, it was shown that there was a significant difference in mortality outcomes in patients of different ages, such as in the IHTSA study which noted a cutoff of age 38 having the greatest difference [7]. The contributing factors to worsened survival and graft function at 5 years after transplant for increasing age are likely multimodal and complex, but also intuitive. Another finding related to age was the increased accuracy of the model for patients with an age > 60 years. Given that a large number of patients needing OHT fall into this age group this is encouraging data that supports use of predictive models. Speculating on why predictability suffers for other age groups below 60 years of age is difficult, but may be due to the effect of different variables or unmeasured variables in younger patients or with congenital heart disease.

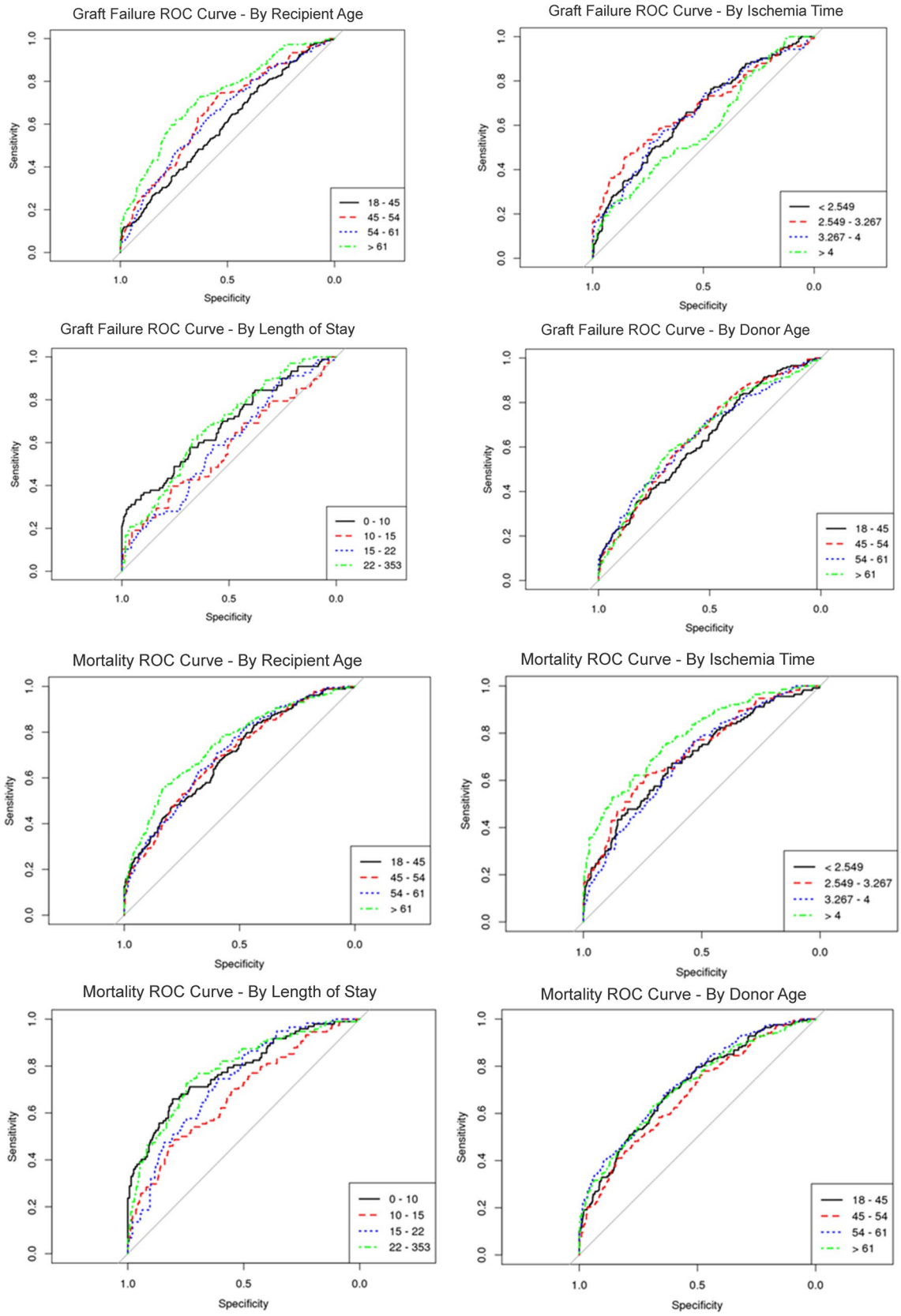
Use of anti-rejection immunosuppressive regimen prior to discharge was the next most influential variable in terms of predicting graft failure after 5 years. However, this was shown to have less of an effect on 5-year mortality prediction than graft failure. The reason for this is unclear, it may be that a more potent regimen of anti-rejection therapy increases the likelihood of graft success but due to toxicity also increases overall mortality, more analysis of this specific finding may be warranted.

Prolonged ischemic time was reported to be significant in our model, which has previously been noted in other models [6, 22–24]. In the present study, prolonged ischemic time was found to have the third and fourth highest weight in predicting 5-year mortality and graft failure, respectively. Prolonged ischemic time has been reported numerous times to worsen short-term graft function and can occasionally lead to short-term graft failure and morbidity [2, 5, 6, 22–24].

The importance of body size was reiterated in our model. Donor BMI and recipient BMI increased the risk of 5-year mortality significantly. Likewise, the risk of graft failure at 5 years was increased by these same variables. This is consistent with previously published studies [6, 7, 25]. No effect from gender differences was seen in our model as opposed to previous models [7, 26].

Lastly, two more recipient variables were found to influence graft failure and mortality prediction at 5 years; pre-transplant creatinine and bilirubin. This is also not surprising, as creatinine and bilirubin are common surrogates for overall kidney and liver function, respectively, it is conceivable that recipients with baseline impaired end organ function may do worse in long-term outcome post-transplant and thus increase rates of mortality and graft failure. The effects of creatinine, BUN to creatinine ratio, and bilirubin have been reported previously in the literature [6, 15, 17].

In terms of generalizability, this model's use of the ISHLT database offers excellent accuracy for a broad range of patient populations. There was no subset of regional


Graft Failure ROC Curve - By Recipient Age
Sensitivity
Specificity
18 - 45
45 - 54
54 - 61
> 61
Graft Failure ROC Curve - By Ischemia Time
< 2.549
2.549 - 3.267
3.267 - 4
> 4
Graft Failure ROC Curve - By Length of Stay
0 - 10
10 - 15
15 - 22
22 - 353
Graft Failure ROC Curve - By Donor Age
Mortality ROC Curve - By Recipient Age
Mortality ROC Curve - By Ischemia Time
Mortality ROC Curve - By Length of Stay
Mortality ROC Curve - By Donor Age

◂**Fig. 2** Subgroup analysis—most influential variables on 5-year mortality and graft failure. Analysis shows that only recipient age > 61 years had a significantly increased AUC for graft failure compared with other age groups ($p < 0.004$). A trend toward increased predictability of 5-year mortality was seen for ischemic time > 4 h compared with other groups ($p < 0.007$). All other subgroup analyses showed no significant differences

data for a specific cohort, but the results may be applied to patients that are demographically similar to the derivation cohort in particular. Previous models that have used this same data set have split validation results according to temporal and regional variation and found similar results across these variations [6, 7].

Similar to the IHTSA algorithm, use of a large number of variables (in this case, 87 precludes easy hand calculation by the clinician and thus computer-based prediction is necessary. This model could potentially be utilized in conjunction with existing donor and recipient registries, such as the United Network for Organ Sharing (UNOS) for optimizing donor and recipient matching for improved 5-year outcomes. In theory, adding a donor to the UNOS registry would automatically enter their many variables and find potential matches that are used to calculate a 5-year mortality and graft failure score using the algorithm. The algorithm could also provide useful information for alterable recipient and donor characteristics that may improve transplant outcomes, such as optimizing variables that affect hospital length of stay as mentioned above.

Limitations

There are several limitations evident in this study. Use of a data registry for retrospective analysis has opportunity for error; missing data, lack of standardization across regions, alterations in standards of care across time eras for example, though this last example is mitigated by the use of data from only between 2005 and 2009. These limitations have been well described [6, 7]. The registry does not collect data of duration of time of recipients on the waiting list or deaths on the waiting list. This data have become less important over time given the improvement in heart failure treatment and left-ventricular assist devices to prolong life prior to OHT. No data were collected on racial differences and its effect on outcomes. One limit of the present study is that there was a significant difference in the level of total bilirubin between the DC and the VC groups with the VC having a significantly higher average level of total bilirubin. Thus, in the VC, the expected outcomes would be worse given that higher bilirubin was shown to significantly influence the risk of mortality and graft failure at 5 years. Despite this, the purpose of this study is to show the predictive value of the machine learning algorithm and although the bilirubin levels in the validation cohort were higher, the algorithm takes this into account when attempting to predict outcomes. One of the benefits of using a machine learning system to predict outcomes is the ability to model non-linear relationships and is not dependent on linear associations to make accurate predictions.

Conclusion

We present an updated and more extensive approach of analyzing and stratifying success of OHT by use of a machine learning algorithm. This model analyzed 87 variables in a non-linear fashion to accurately predict 5-year mortality and graft failure in an individual instance of OHT and provided the top 10 most influential variables for predicting 5-year mortality and graft failure. This model would likely function as a predictive algorithm to estimate risk of 5-year mortality and graft failure in a given donor–recipient match and could also be used to risk stratify several matches on a given donor registry such as UNOS. Use of this model or a model like this could aid the transplant services in confidently selecting matches for transplant that will likely succeed and may increase the number of transplants that are both performed and succeed overall. Identification of patients at a higher risk of mortality and graft failure at the time of discharge can help physicians modify existing post-transplant protocols to emphasize close outpatient clinic follow-up, intensive cardiac rehabilitation, aggressive titration of immunosuppressants, frequent endomyocardial biopsies to evaluate for rejection, emphasize importance of long-term social support, and patient education on medication compliance in an attempt in minimize risk of mortality and graft failure. Given the high relative importance of length of hospital stay in predicting mortality and graft failure post-OHT, we plan to explore and identify predictors of longer length of stay post-OHT in subsequent studies.

Author contributions PA designed the study, extracted the data, drafted the initial manuscript, and critically reviewed the manuscript. SDS drafted the initial manuscript and critically reviewed the manuscript. SA extracted the data and critically reviewed the manuscript. MB analyzed the data, prepared the figures, and critically reviewed the manuscript. MAG analyzed the data, prepared the figures, and critically reviewed the manuscript. JLR critically reviewed the manuscript. BWH critically reviewed the manuscript. PD critically reviewed the manuscript. FM designed the study and critically reviewed the manuscript. RA designed the study and critically reviewed the manuscript. All authors have read and approved the final manuscript.

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Compliance with ethical standards

Conflict of interest The authors have no conflict of interest to declare.

Ethical approval The study complied with the Declaration of Helsinki and was approved by the local institutional review board. The committee waived the requirement for informed consent in consideration of the retrospective nature of the study and anonymous data analyses.

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